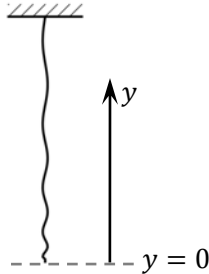


Chap. 47 Sound. The Wave Equation

1. Transverse Wave in Hanging Rope

A uniform rope of length L hangs from the ceiling.



(a) Show that the speed of a transverse wave in the rope is a function of y (the distance from the lower end), and is given by $v = \sqrt{gy}$.

(b) Show that the time it takes a transverse wave to travel the length of the rope is given by $t = 2\sqrt{L/g}$. Note: This time is independent of the actual mass of the rope. Therefore you can determine the length a long rope hanging down from the ceiling by striking its lower end and measuring the time intervals between echoes.

2. Massive Spring (II): Sound Wave in Elastic Medium

In the Lecture, we have learnt the wave equation of sound in the air. In this problem we will deduce the sound wave equation in the elastic medium (solid). Our simple model is a massive spring of the original length L . Its total mass is m (uniformly distributed throughout the spring), and the stiffness (or the force constant) is k .



You can imagine the longitudinal sound wave propagating in the massive spring. Let us label each segment of the spring by its position x at equilibrium, and use $\chi(x, t)$ to denote the displacement of that piece of spring at time t .

(a) Deduce the wave equation for $\chi(x, t)$. Hint: analyze the motion of a little segment of the spring.

(b) Write down the dispersion relation $\omega(q)$ (circular frequency as a function of wave number) for the sound wave. Find the sound velocity c .

Assume a sound wave propagating in the massive spring,

$$\chi(x, t) = \chi_0 \cos q(x - ct).$$

(c) Find the kinetic energy density and potential energy density. Are they always the same everywhere? Note: the energy density here denotes the energy in the spring segment of unit *original* length.

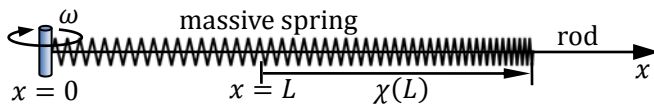
The kinetic energy density and the potential energy density reach their maximum and minimum at the same time and at the same position. It is one of the most important difference between waves and harmonic oscillators, that the kinetic and the potential energy do not mutually transform in waves, but to grow together and to fade together. Does it

contradict with the conservation of energy? No, because what is varying is not the total energy but the energy density. There is no conservation law for energy density, since it can be transported by the energy flow.

(d) What is the energy flow on the spring? Find the instantaneous energy flow as well as the energy flow averaged over a wave period. Note: the energy flow is the amount of energy transferred per unit of time.

3. Massive Spring (III): Dispersive Sound Wave

Let us still look at the uniform massive spring of the original length L , the mass m and the stiffness k . But this time it is threaded on a straight rod that rotates in the plane with constant angular velocity ω about the axis at the end of the rod. The spring is also tied to the axis so that it will not get thrown during the rotation. However the centrifugal force will stretch the spring, so that there could be non-uniform deformation as you can see in the following figure.



We will label each segment of the spring by its equilibrium position x when there is no rotation ($\omega = 0$). Then the relative displacement (deformation) is described by $\chi(x, t)$.

(a) In the rotating frame, what is the additional potential energy that the spring will experience due to the action of centrifugal force?

(b) Deduce the wave equation for the deformation field $\chi(x, t)$. Hint: It might be easier to consider from the energy viewpoint. The total energy of the spring comes from three parts: the kinetic energy, the elastic potential energy and the centrifugal potential energy. Compare with the energy of the non-rotating spring, you might be able to see what extra term should be added to the wave equation.

The wave equation is not a linear equation of χ because $\chi = 0$ is not the equilibrium deformation. So we need to find the static configuration of χ first, upon which we can study the sound wave dynamics.

(c) Under the static condition $\partial_t \chi_0(x, t) = 0$ what is the solution of $\chi_0(x, t)$ that satisfies the wave equation?

(d) How long the spring will be upon rotation with the angular velocity ω ?

(e) What is the critical angular velocity ω_c on which the spring will be ruptured? Note: The rupture strength tends to infinity in our ideal model.

Now we should consider the propagation of the little perturbation $\delta\chi$ on the equilibrium deformation χ_0 .

(f) Substitute $\chi = \chi_0 + \delta\chi$ into the wave equation for χ to obtain the wave equation for $\delta\chi$, from which determine the dispersion relation $\Omega(q)$ of the sound wave in the rotating spring. Note: Let Ω be the circular frequency of the sound wave, and q be the corresponding wave number.

(g) Find the phase velocity and group velocity of the sound. Compare these velocities with the sound speed in the non-rotating massive spring.

In quantum mechanics, the quantized sound wave is called a *phonon*. The dispersion relation

of the sound wave gives the energy-momentum relation of the phonon.

(h) What is the “rest mass” of the phonon in the rotating spring? Note: The so called “rest mass” here is by the analog of sound to light.

Upon rotation, the phonon in the spring acquires rest mass. This is because the rotation breaks the translational symmetry of the spring, which indicates that even a global displacement of the spring will experience the restoring force, and hence the phonon mode will be gapped. The broken symmetry is one possible origin of mass.

Chap. 49 Modes

1. Pulsation Period of White Dwarf

The period of a pulsating variable star may be estimated by considering the star to be executing radial longitudinal pulsations in the fundamental standing wave mode, that is, the radius varies periodically with time, with a displacement antinode at the surface.

(a) Would you expect the center of the star to be a displacement node or antinode?

(b) By analogy with the open organ pipe, show that the period T of pulsation is given by

$$T = \frac{4R}{v}.$$

where R is the equilibrium radius of the star and v is the average sound speed.

(c) Typical white dwarf stars have pressure of 10^{22} Pa, density of $10^{10} \text{ kg} \cdot \text{m}^{-3}$, ratio of specific heats of $4/3$, and radius approximately the same as the radius of the Earth. Estimate the pulsation period of a white dwarf.

2. Wave in Solids

The dispersion of wave in solids can be demonstrated by the simple model of coupled oscillations in one-dimensional crystal. Suppose atoms of mass m are aligned in a line and interact with each other. The interacting force between neighboring atoms can be modeled by a spring of the stiffness k . At equilibrium, these atoms form the one-dimensional crystal with lattice constant a (the spacing between neighboring atoms).



Now if these atoms are excited, sound wave will propagate in the crystal as a collective mode. Let u_l be the displacement of the l th atom from its equilibrium position.

(a) Write down the equation of motion for u_l .

It is known that sound waves are normal modes of these coupled oscillators.

(b) Write down your trial wave $u_l(t)$, and use it to find the frequency of all normal mode.

Show that the sound wave in the crystal is dispersive

$$\omega(q) = 2\sqrt{k/m} \sin|qa/2|,$$

where q is the wave number and ω the frequency.

(c) Find the velocity of sound in the long wave length limit.

(d) Calculate the group velocity. Show that the highest frequency model cannot propagate in the crystal. Can you figure out why?

Chap. 51 Waves

1. Water Wave

Surface wave in liquid has different phase and group velocity. The general expression for phase velocity is

$$v_p = \left(\left(\frac{g\lambda}{2\pi} + \frac{2\pi\sigma}{\rho\lambda} \right) \tanh \frac{2\pi h}{\lambda} \right)^{1/2},$$

where h is the depth, ρ is the density, and σ is the surface tension (for water, $\sigma = 73 \times 10^{-3} \text{ N} \cdot \text{m}^{-1}$).

(a) For deep water, find the phase velocity.

(b) Show that the phase velocity has a minimum when the wave length is $\lambda_0 = 1.7 \text{ cm}$.

According to the wavelength, the surface wave can be classified as gravity wave and capillary wave. The capillary waves are ripples.

(c) Find the phase velocity and group velocity for gravity wave. What is the ratio between group velocity and phase velocity?

(d) Find the phase velocity and group velocity for capillary wave. What is the ratio between group velocity and phase velocity?

(e) If the water is shallow, find the phase velocity and group velocity for gravity wave.